Two-Participant Comparison

Multi-sensor self-monitoring in Dhaka, October 2026: participant 1 (Hemel, days 1–8) and participant 2 (Spriha, days 1–4)

Generated with `pipeline/participants_report.py`, `compare_participants.py`, `relationships.py` and `walk_hydration.py`

1 Summary

Two people wore the same sensor system: pressure-and-motion insoles, an environment node, a wrist band and a smart bottle. Participant 1 (Hemel) was recorded for eight days and participant 2 (Spriha) for four. This report compares them on average and asks which relationships hold for both. It also presents participant 1's test of whether drinking water before walking changes the heart-rate response. The main results are:

- **Sleeping heart rate** is the clearest physiological difference: 70.7 vs 62.3 bpm on average, and every one of her nights (70–72 bpm) is above every one of his (60–64 bpm). Awake heart rate is the same (93 vs 94 bpm).
- **Gait differences are mostly body size**: her strides are 8.8 % shorter in time (1.08 vs 1.18 s; cadence 112 vs 102 steps/min). Her shorter legs alone predict 7.3 %, and after scaling by leg length the difference shrinks to 1.6 %. The two walk in a dynamically similar way.
- **Heart rate follows step rate in both people, to the same degree**: within-day $r = 0.30$ (95 % CI 0.24–0.35) for him and 0.29 (0.16–0.38) for her, with almost the same slope (0.20 and 0.18 bpm per step/min), and positive on all 12 days. This is the one relationship that replicates across people.
- **Water**: she logged a third less per day (1.07 vs 1.57 L; 16 vs 20 mL per kg of body mass), about two-thirds of the EFSA-based amount from drinks for women; he reached about four-fifths of the amount for men.
- **Bedrooms**: hers was much quieter (49 vs 61 dBA) and had fewer particles (63 vs 78 μg/m^3). Indoor and outdoor $PM_{2.5}$ were the same for both (52–56 and 69–71 μg/m^3).
- **Water before walking** (participant 1): at first, walks after more water seemed to raise heart rate less ($\rho = -0.72$). The walks with more water, however, were also the evening walks, and the heart-rate rise is smaller in the evening whatever he drank. A deliberate test on day 8 (500 mL just before a morning walk) gave the largest rise of all. Once the time of day is accounted for, no effect of water remains ($\rho = 0.04$ for water in the 2 h before the walk). The day-8 test is what separated the two explanations.

2 Participants and data

Table 1: The participants and the data available. Leg length is estimated as 0.53 of body height (hip height); it was not measured. BMI: body-mass index.

	Participant 1 (Hemel)	Participant 2 (Spriha)
Sex	male	female
Height	1.71 m (5 ft 7.5 in)	1.47 m (4 ft 10 in)
Body mass	77 kg	68 kg
BMI	26.2 kg/m^2	31.3 kg/m^2
Estimated leg length	0.91 m	0.78 m
Days recorded	8	4
Steady strides analysed	16,161	927
Nights with band sleep	8 (6.5 h on average)	3 (only part of each night)
Nights with the bedroom node	6	2
Days with a drinking log	7	3
Band heart-rate sampling	every minute	every 10 min (days 1–2), every minute (days 3–4)
Typical travel	walking, metro, rickshaw	rickshaw

Some measures are not comparable between the two, and the report does not compare them. Her band recorded only part of each night (2.0–3.5 h), so sleep duration and stages are not compared. Her band was also off for long stretches on some days, so daily step totals are not compared (5,900 vs 1,600). The environment node's self-heating was larger in her recordings (warm-up of 10–12 °C after a cold start, against a correction of 5 °C), so her rooms' temperatures, which read 4–7 °C warmer than his, are partly an artefact. On her day 4 the insoles were deliberately not worn, so her gait comes from days 1–3.

3 Day-level averages

- **Heart rate asleep**: her nights are 8–10 bpm higher, with no overlap between the two people (Fig. 1a). Sex, fitness and body composition all affect resting heart rate, and two people cannot separate them. The difference is still large and very stable from night to night (SD 1–2 bpm), so it is a reliable individual baseline for the digital twin.
- **Heart rate awake** is the same on average, but his days range more widely (86–104 bpm) because they included metro trips and long walks;

Table 2: Per-day means ± SD over days (number of days). Comparable: *yes* = measured in the same way for both; *partly* = comparable but dependent on what each person did; *no* = a measurement difference dominates.

Measure	Hemel	Spriha	Diff.	Comp.
HR asleep (bpm)	62.3 ± 1.7 (8)	70.7 ± 0.9 (3)	+8.4	yes
Lowest 10-min HR asleep (bpm)	56.6 ± 1.4 (8)	66.6 ± 1.4 (3)	+10.0	yes
HR awake (bpm)	94.3 ± 5.8 (8)	92.7 ± 4.3 (4)	−1.6	yes
Highest HR of the day (bpm)	145 ± 12 (8)	120 ± 13 (4)	−25	partly
Walking HR (bpm)	113 ± 12 (6)	105 ± 5 (3)	−8	partly
Cadence (steps/min)	102.1 ± 3.5 (8)	111.3 ± 5.8 (3)	+9.2	yes
Water (L/day)	1.57 ± 0.44 (7)	1.07 ± 0.48 (3)	−0.50	yes
per kg body mass (mL/kg)	20.4	15.7	−4.7	yes
Bedroom $PM_{2.5}$ ($\mu g/m^3$)	77.7 ± 14.6 (6)	63.0 ± 8.5 (2)	−14.7	yes
Bedroom sound (dBA)	60.7 ± 4.2 (6)	48.6 ± 1.1 (2)	−12.1	yes
Indoor $PM_{2.5}$ ($\mu g/m^3$)	51.5 ± 8.8 (6)	56.0 ± 20.1 (3)	+4.5	yes
Outdoor $PM_{2.5}$ ($\mu g/m^3$)	69.3 ± 14.0 (8)	70.7 ± 11.2 (3)	+1.4	yes
Indoor CO_2 (ppm)	652 ± 130 (6)	427 ± 128 (3)	−225	partly
Sleep recorded (h)	6.5 ± 0.7 (8)	2.9 ± 0.8 (3)		no
Steps (band)	5,879 ± 2,002 (8)	1,620 ± 1,528 (4)		no
Indoor temperature (°C)	25.4 ± 2.9 (6)	32.4 ± 2.9 (3)		no

his highest heart rate of the day was correspondingly higher (145 vs 120 bpm). Her walking heart rate is lower (105 vs 113 bpm), but her walks lasted only 2–4 minutes.

- **Water**: her three logged days (0.60, 1.08 and 1.55 L) are all below his average, though her latest is within his range of 0.89–2.16 L (Fig. 1c). Measured against about 80 % of the EFSA adequate total water intake (the part expected from drinks: 2.0 L for men, 1.6 L for women), he reached 78 % on average and she 67 %. On her days 2–3 most of her water came at night or in the evening.

- **Bedrooms**: her bedroom was 12 dBA quieter, which is about a halving of loudness, and cleaner. Both are above the WHO 24-hour $PM_{2.5}$ guideline (15 $\mu g/m^3$) all night.

- **CO_2**: indoor CO_2 was lower in her recordings. The sensor reads below the outdoor level at times, so only changes within a day are reliable; both showed build-ups in closed spaces (his to 1,500–2,500 ppm in the metro, hers to 800 ppm in a closed evening room).

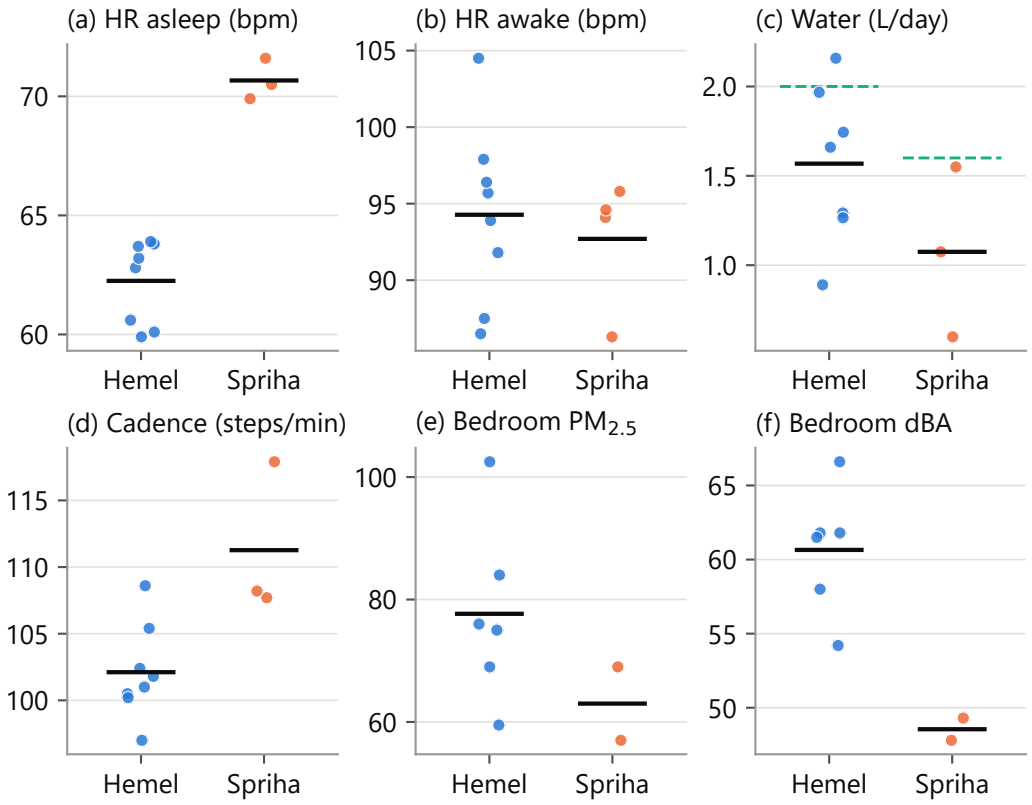


Figure 1: Day-level measures of the two participants: each dot is one day, the black bar is the mean over days. (a) Mean heart rate asleep, (b) mean heart rate awake, (c) fluid from the smart-bottle log (dashed: about 80 % of the EFSA adequate total water intake, i.e. 2.0 L for men and 1.6 L for women), (d) walking cadence, (e) bedroom $PM_{2.5}$ and (f) bedroom sound level (medians of the overnight recording; $PM_{2.5}$ in $\mu g/m^3$).

4 Gait and body size

Her cadence is higher and her strides shorter in time (Fig. 2). For a pendulum-like leg, the natural stride time scales with $\sqrt{L/g}$, where L is leg length and g is gravity. Her legs are an estimated 14 % shorter, which predicts strides $1 - \sqrt{0.78/0.91} = 7.3\,\%$ shorter in time. The measured difference is 8.8 % (1.076 vs 1.180 s, medians). Divided by $\sqrt{L/g}$, the stride times are 3.81 and 3.88 (Fig. 3), only 1.6 % apart. About five-sixths of the difference is therefore explained by height, and the two walk in a dynamically similar way. Stance from the left pressure sensors (original and healthy in both) is the same, 64 %

of the stride.

This matters for the digital twin. A gait model built on one person transfers to the other more easily if cadence and stride time are expressed relative to body size. With leg length estimated rather than measured, the remaining 1.6 % is within the uncertainty of the estimate.

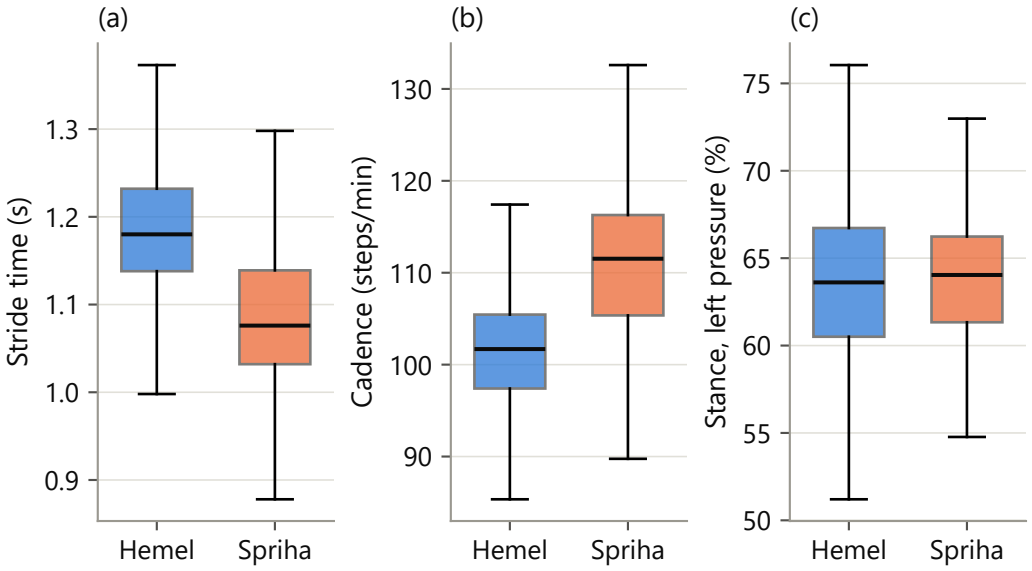


Figure 2: Gait of the two participants in all walks (steady strides; Hemel: days 1–8, 16,161 strides; Spriha: days 1–3, 927 strides). (a) Stride time, (b) cadence and (c) stance phase from the left insole's pressure sensors (the left insole's sensors were original and healthy for both participants).

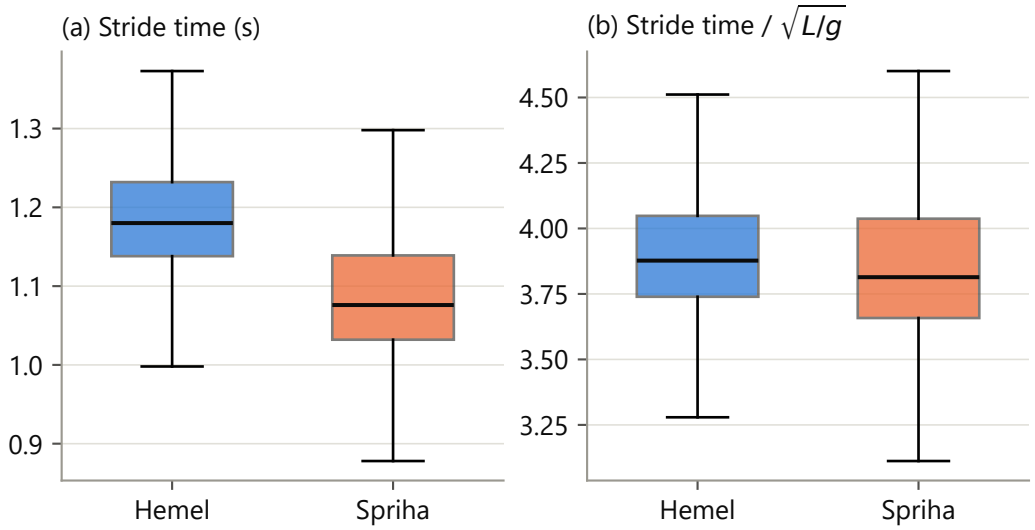


Figure 3: Stride time of all steady strides (Hemel 16,161, Spriha 927). (a) As measured: Spriha's strides are 9 % shorter in time. (b) Divided by $\sqrt{L/g}$, the natural time scale of a pendulum-like leg of length L (taken as 0.53 of body height; $g = 9.81\,\mathrm{m/s^2}$). After this scaling the difference is 1.6 %.

5 Settings

By setting (Fig. 4), the two had similar indoor and outdoor particle and sound levels. Her bedroom was quieter and had fewer particles, and her rickshaw rides had more particles than his (median 74–98 vs 53–56 $\mu\mathrm{g/m^3}$ on rickshaw rides). Heart rate in vehicles was lower for her (113 vs 137 bpm), but his figure is dominated by one metro evening. Temperatures are shown but not interpreted (Section 2).

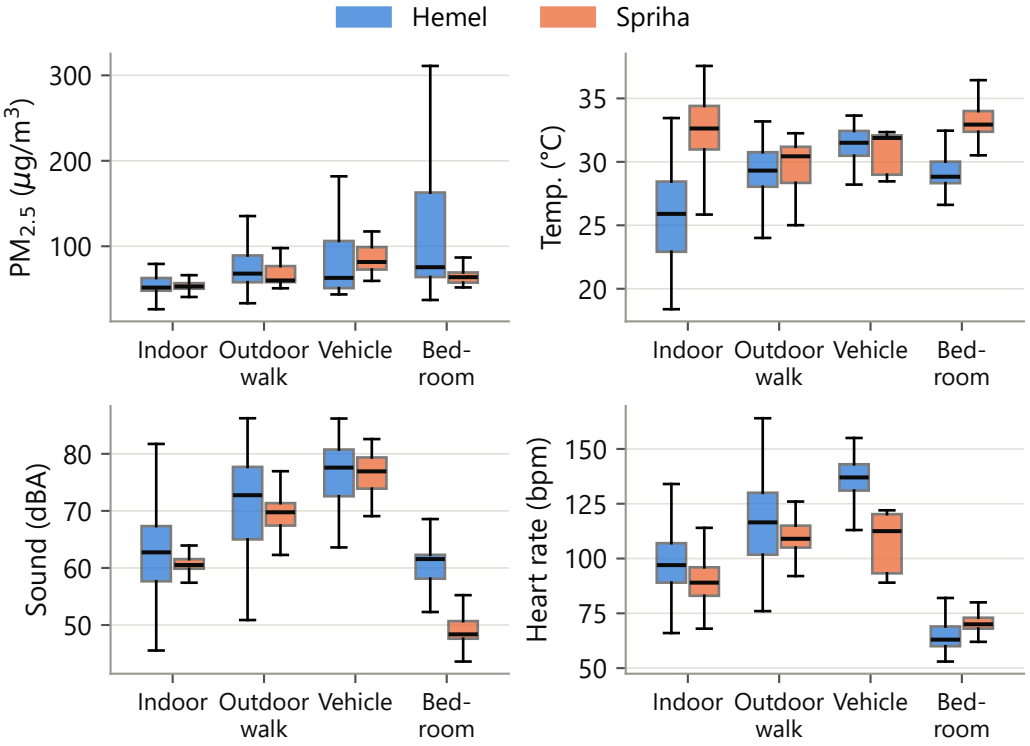


Figure 4: Environment and heart rate by setting for the two participants (1-minute values; Hemel days 1–8; Spriha days 1–4). Temperatures include the −5 °C correction. Spriha's band recorded heart rate every 10 minutes on days 1–2 and every minute from day 3, so her heart-rate boxes rest on fewer minutes.

6 Correlations that can be compared

Every relationship below is computed in the same way for both participants. Each variable is first centred on its own day mean, so differences between days cannot create a correlation. Confidence intervals come from a bootstrap over 10-minute blocks, because neighbouring minutes are not independent.

1. **Heart rate and step rate: the relationship that replicates.** The strength (r = 0.30 and 0.29) and the slope (0.20 and 0.18 bpm per step/min, i.e. about +19 bpm at 100 steps/min) are almost identical in two people of different sex and size (Figs. 5 and 6). Median heart rate rises from 93 and 91 bpm in minutes without steps to 112 and 107 bpm in minutes with more than 90 steps. The correlation is positive on all 12 days (Fig. 6b). It explains about 9 % of the minute-to-minute variation

Table 3: Relationships measured for both participants: within-day Pearson r (95 % CI); *CI excludes zero. Minutes: heart rate–steps 6,792 (Hemel) and 832 (Spriha); asleep 2,362 and 326; environment 4,328 and 1,612. Slopes (Hemel / Spriha): 0.20 / 0.18 bpm per step/min; −1.06 / −0.49 bpm per dBA; 1.94 / 0.31 bpm per °C; RH 0.62 / −0.91 % per °C.

Pair	Hemel (days 1–8)	Spriha (days 1–4)	Reading
HR vs steps, awake	.30* (.24, .35)	.29* (.16, .38)	replicates
HR vs sound, asleep	−.23* (−.32, −.14)	−.22* (−.38, −.04)	shared trend
HR vs temperature, asleep	.24* (.16, .33)	.12 (−.05, .29)	participant 1 only
HR vs $PM_{2.5}$, asleep	−.04 (−.12, .04)	−.14 (−.35, .05)	none
HR vs CO_2, asleep	−.09 (−.17, .00)	.06 (−.06, .17)	none
PM_{10} vs $PM_{2.5}$	1.00*	.98* (.97, .98)	sensor check
RH vs temperature	.22* (.11, .32)	−.56* (−.74, −.35)	opposite

in heart rate, and it holds only for sustained activity: minute-to-minute changes are not related ($r = -0.03$). Heart rate responds to activity over several minutes rather than within the same minute.

2. **Heart rate and sound during sleep: shared, but probably not causal.** Both show $r \approx -0.23$: quieter minutes have higher heart rate. In neither person are the minute-to-minute changes related ($r = 0.01$ and 0.00), so the association comes from slow trends across the night that both variables share (for example a fan switched on, or deeper sleep later in the night), not from sound changing heart rate.

3. **Heart rate and bedroom temperature** is positive for participant 1 (+1.9 bpm per °C, CI excludes zero) but not significant for participant 2, whose bedroom temperature is distorted by the node's warm-up. This fits the known effect of a warm room on sleeping heart rate, but it is shown by one person only.

4. **PM_{10} and $PM_{2.5}$** ($r = 0.98$–1.00) come from one optical sensor. This is a check that the sensor behaves the same in both homes, not a finding.

5. **Humidity and temperature have opposite signs.** In his recordings they rise together (a weather and room effect). In hers, relative humidity falls as temperature rises ($r = -0.56$; −0.97 on her day-4 night). That is the signature of the sensor heating the air it measures, so it belongs to the sensor-system results.

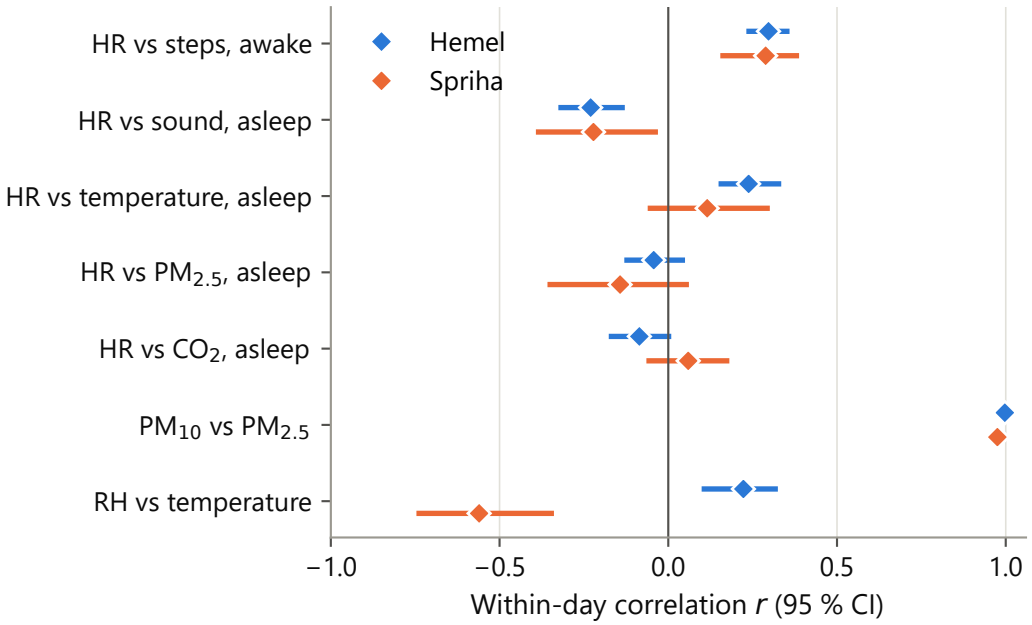


Figure 5: The relationships measured for both participants: pooled within-day Pearson r (each variable centred on its own day mean) with 95 % CI from a 10-minute block bootstrap (Hemel days 1–8, Spriha days 1–4; sleep pairs: nights with both band sleep and the bedroom node).

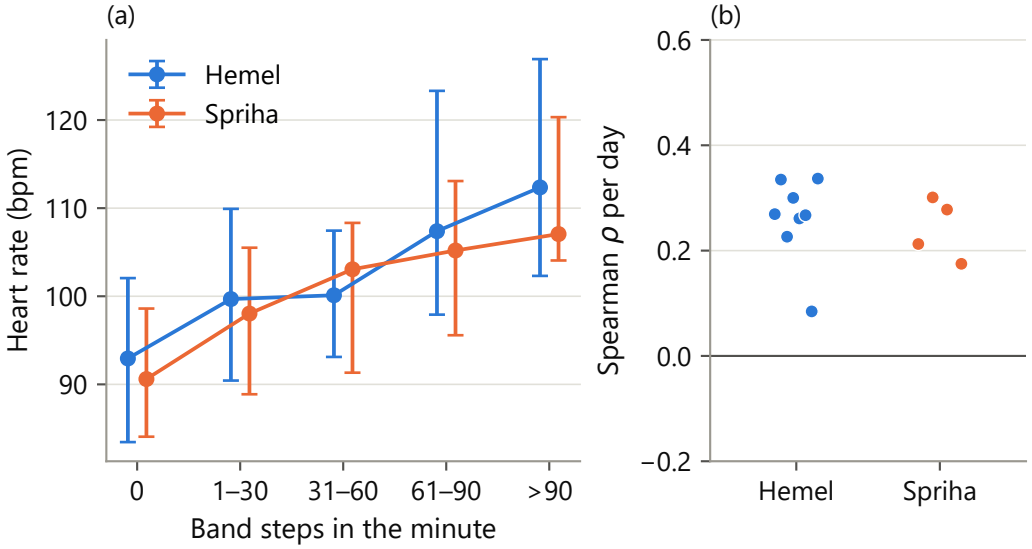


Figure 6: Heart rate and step rate while awake. (a) Median heart rate (bars: inter-quartile range) by the number of band steps in the minute, after removing each day's mean so that day-to-day differences do not add to the curve (bins with fewer than 10 minutes are omitted). (b) The per-day Spearman correlation; every day of both participants is positive.

7 Participant 1: water before walking and heart rate

Question. Does drinking water before a walk lower the heart-rate response to walking? The response is measured as the rise of heart rate during the walk above the resting value 5–30 minutes before it. Eleven walks over days 1–8 had a valid heart rate. Two further walks were excluded because the band mostly reported step rate instead of pulse (cadence lock).

What the first seven days suggested. Walks after more water had smaller rises. Over days 1–7 the rise correlated with water drunk earlier that day at $\rho = -0.76$ and with water in the 2 h before at $\rho = -0.61$ (Table 4). On day 7 he drank before both walks, and the evening walk had the lowest walking heart rate of the study (92 bpm).

The competing explanation. Water accumulates over the day ($\rho = 0.79$ between water so far and the hour), and the heart-rate rise is smaller later in the day ($\rho = -0.69$) because his resting heart rate in the evening is already higher (93–138 bpm vs 81–108 bpm before morning walks). On days 1–7 every walk after more than 700 mL that day was an evening walk (Fig. 7a), so water and time of day could not be told apart.

The test. On day 8 he drank 500 mL immediately before a *morning* walk, breaking the link between water and time of day. If water lowered the response, this walk should have had a small rise. It had the largest of all eleven walks (+37 bpm; walking heart rate 123 bpm; Fig. 9).

Table 4: Heart-rate rise when walking against water drunk, before and after the day-8 test (Spearman ρ; partial: after removing the time of day from both variables). 95 % bootstrap CIs for days 1–8: rise vs water earlier that day −0.97 to −0.24; its partial −0.95 to 0.48.

	Days 1–7 (10 walks)	Days 1–8 (11 walks)
Rise vs water in the 2 h before	−0.61	−0.31
partial, time of day removed	−0.34	0.04
Rise vs water earlier that day	−0.76	−0.72
partial, time of day removed	−0.48	−0.39
Rise vs hour of the walk	−0.71	−0.69

Result. With time of day accounted for, water in the 2 hours before a walk has no relationship with the heart-rate response (partial $\rho = 0.04$). For water drunk earlier in the day, a weak negative tendency remains (partial $\rho = -0.39$), but its confidence interval runs from −0.95 to +0.48 with eleven

walks. Walking heart rate itself is not related to water ($\rho = 0.20$), and a model adjusted for temperature gives -0.1 bpm per 100 mL (95 % CI -5.3 to $+6.2$).

Why this is a useful finding. The analysis shows how a convincing pattern in self-tracking data can come entirely from daily routine, here drinking more and walking later in the evening. A single planned walk that broke the routine was enough to separate the two explanations. For the digital twin, time of day must be modelled before any behaviour is credited with an effect on heart rate. To test hydration properly, the next step would be to vary water deliberately (for example 0 vs 500 mL) before walks at the *same* time of day on different days.

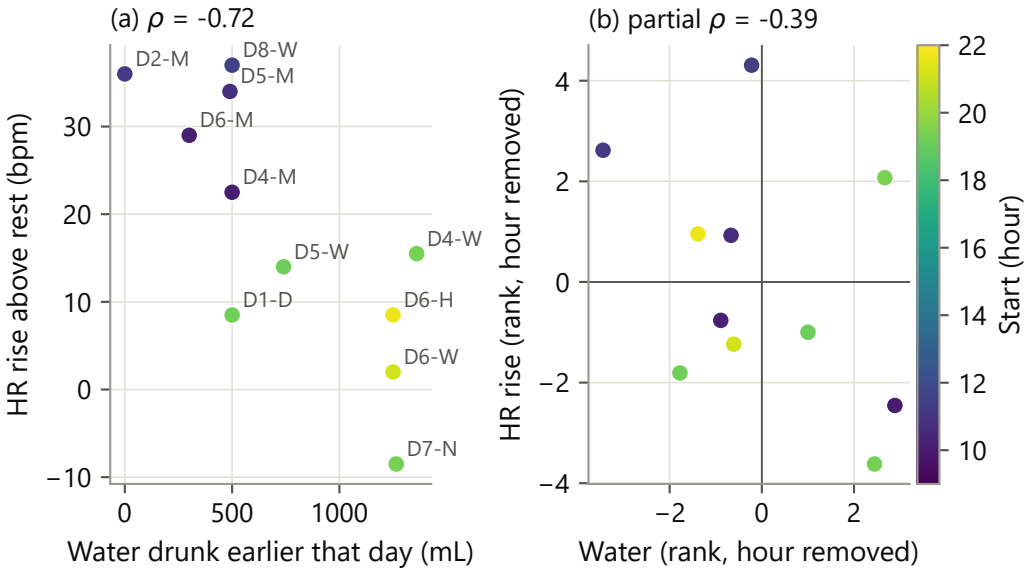


Figure 7: Participant 1: water drunk earlier in the day and the rise of heart rate when walking (11 walks, days 1–8; rise = walking heart rate minus the resting heart rate 5–30 min before). Colour: hour the walk started. (a) Walks after more water had smaller rises ($\rho = -0.72$, 95 % CI -0.97 to -0.24), but the rise also shrinks later in the day ($\rho = -0.69$) and water accumulates through the day ($\rho = 0.79$). (b) Ranks of both after removing the time of day: partial $\rho = -0.39$ (95 % CI -0.95 to 0.48).

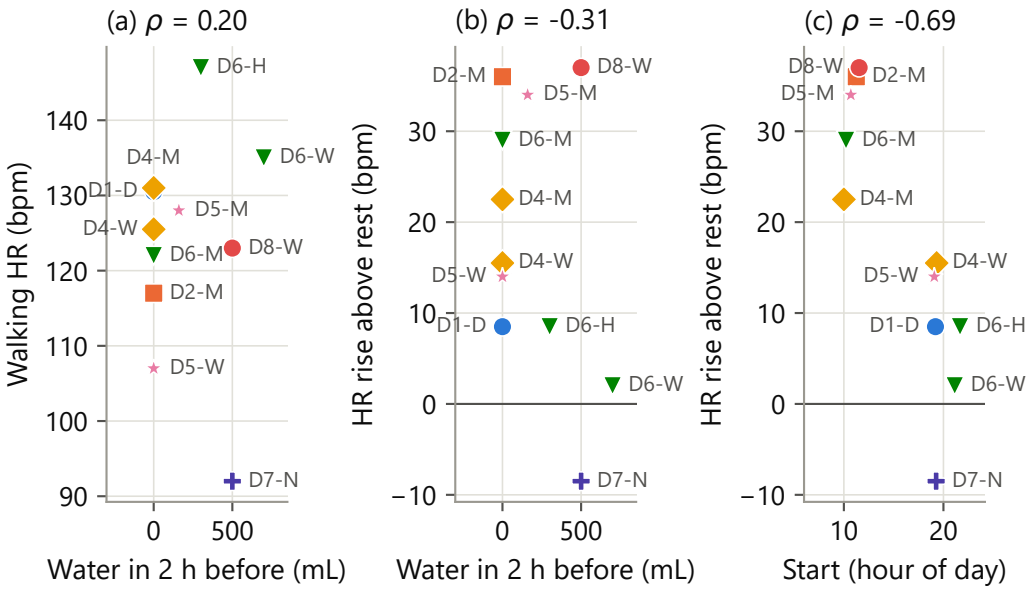


Figure 8: Water drunk before walking and heart rate during the walk (11 walks over days 1–8; median heart rate of walking minutes excluding suspected cadence lock). Walks D3-M, D6-E are excluded because most of their walking minutes were suspected cadence lock. (a) Walking heart rate against fluid from the bottle log in the 2 hours before the walk; (b) rise of heart rate above the resting value 5–30 minutes before the walk, against the same; (c) that rise against the time of day. Spearman ρ with bootstrap 95 % CI: (a) 0.20 (-0.58 to 0.79), (b) -0.31 (-0.86 to 0.60), (c) -0.69 (-0.86 to -0.23); (b) after allowing for the time of day: partial $\rho = 0.04$. Labels: day and walk.

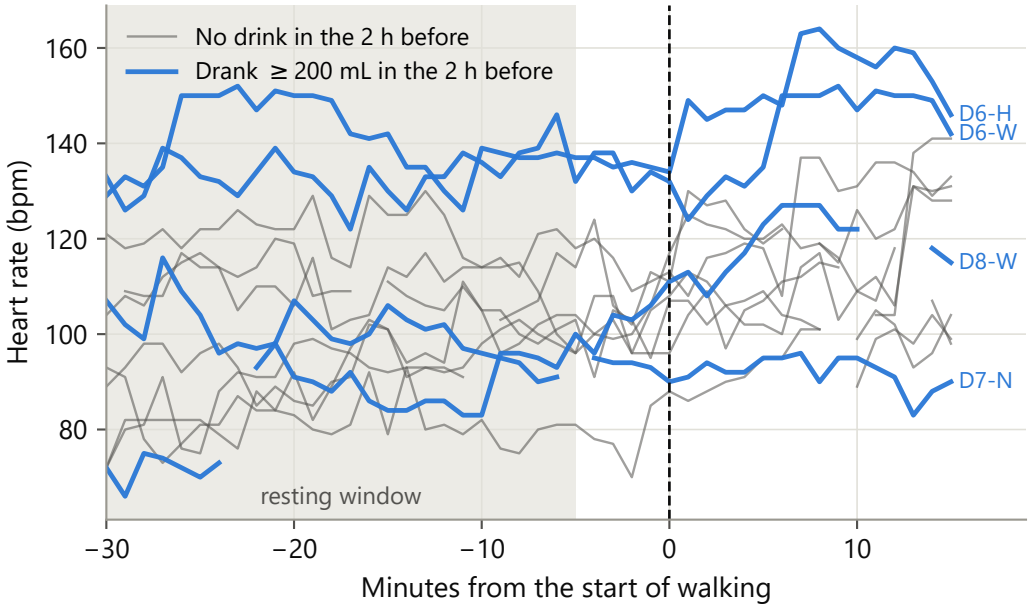


Figure 9: Wrist heart rate around the start of every walk with a valid walking heart rate (11 walks, days 1–8), aligned at the first walking minute (dashed). Blue: walks with at least 200 mL drunk in the previous 2 hours (labelled); grey: the others. The shaded band is the resting window used for the rise of heart rate above rest.

8 Limitations

- Two people, 8 and 4 days: the comparison describes these two individuals. Differences between them cannot be attributed to sex, body size or fitness.
- Her sleep and step data are incomplete because the band was worn only part of the time, and her heart rate was sampled every 10 minutes on days 1–2.
- Leg length is estimated from height ($0.53 \times$ height), not measured.
- Temperature is affected by the node's self-heating, which differs between recordings. Absolute humidity and within-recording changes are reliable; absolute temperatures should be compared only after recalibration.
- The water-before-walking analysis rests on eleven walks of one person, with the drinking times taken from the bottle log (one time on day 8 entered as an estimate).